Emna KAANICHE

Software Engineering Student — AI & Full Stack Development
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SUMMARY

From graduating top of my undergraduate class to pursuing Software Engineering at ENIS, my background combines strong scientific rigor with a passion for building scalable software systems. Specializing in **Artificial Intelligence** and **Full Stack Development**, I design robust, end-to-end solutions ranging from intelligent data pipelines to modular production applications.

EDUCATION

National School of Engineers of Sfax (ENIS) **2024 – Present**
Engineering Degree / M.S. in Computer Science & Software Engineering
2nd Year: High Honors (GPA: 15.25/20) — 1st Year: High Honors (GPA: 14.64/20)

FSEGS & University of Le Mans (Dual Degree) **2021 – 2024**
B.S. in Business Information Systems
Ranked 1st out of 282 students (Valedictorian, Class of 2021–2024)

WORK EXPERIENCE

AI Developer Intern — Intelligent Support Response Agent **June – Aug. 2026**
Summer Internship — CLINISYS • On-site
Keywords: Hybrid RAG, Qdrant, BM25, Cross-Encoder, spaCy, LLMs, ERP
- Engineered an autonomous AI agent integrated into the CliniSYS medical ERP support platform, automating ticket analysis, knowledge retrieval, and response drafting in **11s** (down from a 16h human average).
- Curated a 775 Q&A pair knowledge base (33 categories) extracted from ERP source code; built a hybrid RAG pipeline with cross-encoder reranking and an abstention threshold (0.60) for automated human escalation.
- Implemented local PII redaction using spaCy prior to cloud LLM processing and deployed 3 defensive guardrails (relevance scoring, prompt injection defense, API failover).

Full Stack Developer Intern — SOAP to REST Architecture Migration **Feb. – May 2024**
Capstone Internship (PFE) — CLINISYS • On-site
Keywords: Spring Boot, React, Redis, RESTful APIs, Swagger, SOAP
- Migrated legacy SOAP services to a modern RESTful architecture using a modular Spring Boot backend, a redesigned React frontend, Redis caching, and Swagger documentation, accelerating response times by over **2x**.

ACADEMIC & RESEARCH PROJECTS

AI Research Project — Project VORTEX (Automated Test Generation) **Jan. – Apr. 2026**
Team Capstone Project (3 members) — ENIS • On-site
Keywords: Multi-Agent Systems, LangGraph, LangChain, ChromaDB, Groq LLM, RAG, ICAART 2027 Submission
- Co-architected a multi-agent system designed to automatically translate User Stories into executable software test suites using Gherkin syntax.
- Developed the agent orchestration pipeline (LangGraph + Groq LLM) and integrated a hybrid RAG system with feedback loopbacks, achieving **60–80%** code coverage on Spring Boot microservices.
- Co-authored and submitted a peer-reviewed research paper presenting the system architecture and experimental results to the **ICAART 2027** international conference.

AI, Computer Vision & Machine Learning Projects **2026**
Keywords: Python, OpenCV, MediaPipe, Deep Learning, Scikit-Learn, FastAPI, Docker
- **Real-Time Sign Language Recognition (CV & DL) :** Engineered a real-time gesture pipeline extracting hand landmarks (OpenCV, MediaPipe), trained deep learning models for gesture classification, and exposed inference endpoints via a FastAPI REST API.
- **Customer Segmentation & Sales Forecasting (ML & Full Stack) :** Performed exploratory data analysis and feature engineering (RFM), built unsupervised clustering models (K-Means), developed sales forecasting regression models, and containerized the solution with Docker.

SKILLS

- **AI & Data Science**: Machine Learning (Scikit-Learn, XGBoost), Deep Learning (PyTorch, TensorFlow), Computer Vision (OpenCV, MediaPipe), Generative AI & LLMs, Hybrid RAG (Qdrant, ChromaDB, BM25, Reranking), Multi-Agent Systems (LangGraph, LangChain), NLP (spaCy), Prompt Engineering
- **Software Engineering & Full Stack**: Python, Java, Spring Boot, FastAPI, React, Flutter, Dart, JavaScript, Microservices Architecture, Design Patterns, REST APIs, Asynchronous Programming, JWT
- **Databases & DevOps**: PostgreSQL, MySQL, Redis, Docker, Git, GitHub, Swagger, Unit Testing (JUnit, Cucumber)
- **Soft Skills**: Autonomy & Self-Assessment, Continuous Learning, Teamwork, Adaptability, Rigor & Quality Assurance

CERTIFICATIONS

- Certification: **Talking to AI: Prompt Engineering for Project Managers** — Project Management Institute (PMI).
- Certification: **AI Capabilities and Limitations** — Anthropic Education.

LEADERSHIP & INTERESTS

- **Microsoft Tech Club FSEGS:** President (2022–2023), Lead Designer (2023–2024) — *Spearheaded strategic restructuring and turnaround of an inactive tech club, reviving student engagement and technical workshops.*
- **Pitch Yourself For A Career:** Music Director / Manager (2025–2026), Event Organizer (2024–2025).
- **Conservatory of Music:** Pursuing formal music theory and practical studies (Oud performance) alongside engineering studies.

LANGUAGES

Arabic: Native — **French**: Professional Working Proficiency / Fluent — **English**: Professional Working Proficiency / Fluent